

Redirected Walking in Overlapping Rooms

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ABSTRACT

Walking in larger virtual environments than the physical one can lead to collisions with physical boundaries. Multiple locomotion techniques like Redirected Walking (RDW) and Overlapping Architecture (OA) aim to overcome this limitation. Combining these two has yet to be investigated in large physical spaces with resets. In this work, a hybrid locomotion method was implemented that combines RDW and OA. A user study was conducted where participants collected items in a virtual environment with multiple rooms. The study showed that the distance walked between resets was increased substantially, thus showing the solid advantages of combining OA and RDW.

CCS CONCEPTS

• **Human-centered computing** → *Virtual reality*.

KEYWORDS

walking, redirected walking, overlapping architecture

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1 INTRODUCTION

Locomotion is crucial for Virtual Reality (VR) applications requiring users to navigate in a virtual environment (VE). Real walking is generally seen as the gold standard in terms of immersion and presence. However, walking in a VE larger than the available tracking space leads to collisions with the boundaries. Thus, redirection techniques were introduced to circumvent this issue by increasing the distances the user can navigate before being too close to the boundaries.

These techniques still involve the users physically walking but use various methods to keep them inside the tracking space when

navigating a large VE. Two examples are Redirected Walking (RDW) and Overlapping Architecture (OA).

RDW, first described by [Razzaque et al. 2001], aims to reduce the large space required for real Walking by exploiting human's visuo-haptic dominance. RDW induces a sensory mismatch, resulting in a deviation of the virtual from the real walking trajectory. RDW is typically combined with resetting [Williams et al. 2007], which is exclusively used to prevent an imminent user's collision near physical boundaries or obstacles. Once a user is approaching boundaries of the tracking space, the system displays a message asking the user to stop walking and start turning around. While turning, a rotational gain beyond detection thresholds is applied until their heading points away from the obstacle or boundaries, from where the virtual exploration continues.

"Impossible spaces" introduced by Suma et al. [Suma et al. 2012] strategically map portions of the VE to the physical environment (PE), where multiple virtual locations can occupy the same physical space. OA is part of this redirection technique and describes overlapping virtual rooms in the tracking space allowing to increase the virtual space. This effect can be subtle if the overlap is less than 55.57% for small $3.6m \times 7.3m$ and 31.06% for large $9.1m \times 9.1m$ rooms.

Only limited research exists on the combination of RDW and OA [Langbehn et al. 2018; Suma et al. 2012]. Both studies' implementations used unique, purposefully designed VEs and did not consider resets since the user was always inside the tracking space. Moreover, in both studies, RDW was only used when the user walked between the rooms. It is disabled as soon as they enter one. Combining RDW with OA techniques has yet to be investigated in large PEs where resets are also used. We hypothesize that combining the methods yields better results regarding the walked distance between resets in contrast to applying the techniques mutually exclusively and that such a combination benefits exploring large VEs. Our work fills the research gap by implementing a hybrid redirection technique combining RDW, including resets, with OA.

2 SYSTEM OVERVIEW

Our VE was designed in Unity. The VE consists of a central corridor with three branching corridors (see Figure 1). The two outer corridors have doors to four rooms on one side, and the middle has doors to four rooms on both sides. In total, there are 16 rooms. The user's starting position is at the junction of the central corridor and the middle corridor shown by the magenta dot in the image. From there, they can freely explore the virtual rooms by approaching

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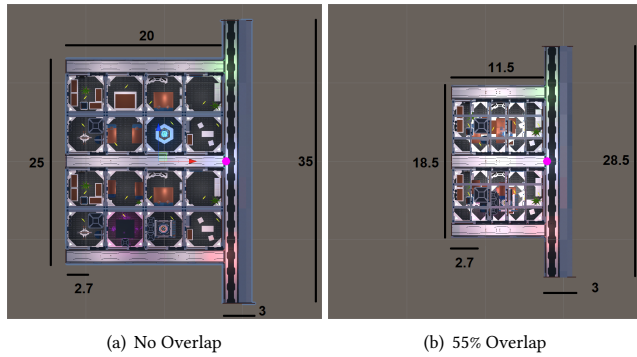


Figure 1: Illustration of the virtual environment

the automatic sliding doors. Whenever the user enters one, the other walls from other rooms need to be hidden, thus preventing the user from seeing the overlap. Only the room the user is entering is enabled, while the others are disabled, i.e., not rendered and deactivated from the game logic. The sliding doors ensure this will happen before the door opens.

As shown in Figure 1, the rooms are spatially overlapping in both direction of the corridors. The shift is towards the start position.

The gains for the RDW algorithm were set to rather conservative values less than the ones used in literature as follows: the translation gain was set to ± 0.22 , the rotation gain was set to ± 0.15 , and the curvature gain was set to a value corresponding to a 22 m radius. Moreover, following the implementation of the RDW toolkit, the steering algorithm was set to Steer-to-Center.

3 USER STUDY

The user study employed a mixed design, consisting of both between-subject and within-subject factors. Twenty-six participants were recruited for the study, with 14 assigned to the RDW + reset condition and 12 assigned to the reset-only condition. Each participant experienced the virtual environment (VE) twice: once with an overlap of 55% and once without. Both groups were exposed to reset-to-center mechanisms to prevent collisions at the boundaries. The order of the trials was randomized, ensuring that half of the participants started without the OA, while the other half started with it. Importantly, no participants were excluded from the study due to simulator sickness.

The study starts with the standardized simulator sickness questionnaire. Afterwards, two ten-minute trials, one with the OA and one without it, are performed in VR. The participant is asked to collect 16 stars distributed throughout the environment, one in each room. The trial ends either when the participant collects all-stars or the 10-minute time limit is exceeded. Finally, the participant has to complete the SSQ questionnaire again with an addition of the SUS questionnaire and some questions regarding the noticeability of the introduced overlaps.

Generally, the results from the OA questionnaire indicate that participants noticed a difference with regard to the “no overlap” condition (19/26). From these the 19, 13 pointed out that something was off with the layout and rooms could have been overlapping. In

the final OA question, 9 correctly identified the trial with the OA. This means that only 34% can manage to detect the overlap and correctly identify when it occurs. No user provided negative feedback about the combination of OA and RDW. Many reported that it is not disruptive at all. Some even explicitly said they preferred it since they had fewer resets. This means that despite user detecting overlap it doesn’t negatively affect their experience. Moreover, a high usability score of 77 was found which supports that the system was easy to use.

For this study, d_{Reset} was considered as an objective measurement which is the average distance between resets. Notably, RDW improves this distance for the two main conditions. In addition, a remarkable improvement was observed in the overlap condition, specifically in the RDW+reset condition, which provides further validation of our assumption regarding the usefulness of combining redirection methods.

	Overlap		No Overlap	
	Reset Only	RDW + Reset	Reset Only	RDW + Reset
d_{Reset}	$5.9 \pm 0.6m$	$6.8 \pm 0.88m$	$5.9 \pm 0.43m$	$6.1 \pm 0.5m$

4 CONCLUSION AND FUTURE WORK

In this study, we investigated the combined usage of OA and RDW. A user study was conducted where a collection task was implemented. The participants were divided into two conditions: one experiencing continuous RDW and one only encountering resets. The results suggest continuous RDW is compatible with overlapping architecture and should therefore be investigated further. The hybrid approach was well received by the users through standardized questionnaires. A future, more extensive study is planned that would allow for a statistical analysis of the results. Likely, a more sophisticated RDW algorithm could be explored based on artificial potential fields [Hirt et al. 2019]. Further, the OA could be combined with spatial alignment [Williams et al. 2021]. Here, it would be possible to redirect a user so that the rooms would be aligned with the physical tracking space. Moreover, many different redirection techniques other than RDW such as change blindness still need to be evaluated in combination with OA.

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